



UX APPROACH DESIGN FOR ELECTRONICS ENGINEERS

Affiliation Disclosure

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Introduction

In the past few years of taking on students from different programs the realization of Academe Mismatch with the skills set required by future employees of companies are not met by the current education programs. This is a consistent problem across all courses. The issue is most likely a problem with educational setup we follow that is based on what CHED requires rather than what the industry requires.

This material focuses on the basic material not covered by the CHED sanctioned materials but nonetheless useful to make the possibility of transitioning from school to work possible. This is the gap that a properly set up internship program can address.

Difference between Research and Research and Development

What is Research and Development ?

- Broken down into two distinct concepts
 - Research (investigationem)
 - Development (progressionem)

What is Research and Development ?

- Broken down into two distinct concepts
 - Research (investigationem)
 - Systematic Inquiry to describe, explain, predict and control the observed phenomenon.
 - Earl Robert Babbie

What is Research and Development?

- Broken down into two distinct concepts
 - Research (investigationem)
 - Research can encompass
 - Human Behavior
 - Machine Behavior
 - Physical Phenomenon
 - Human Activity
 - Human Anatomy, Physiology and Functional Systems
 - Behavior of Matter
 - Properties of Matter

What is Research and Development?

- Research is
 - Open in its approach
 - Open Ended
 - Systematic
 - Tries to find out what is there, why it is this way and what contributes to it /disrupts it.

V. Dupro

What is Research and Development?

- Research

- Doesn't care if it becomes a product or service it just wants to answer questions using a systematic manner to arrive at the end goal.

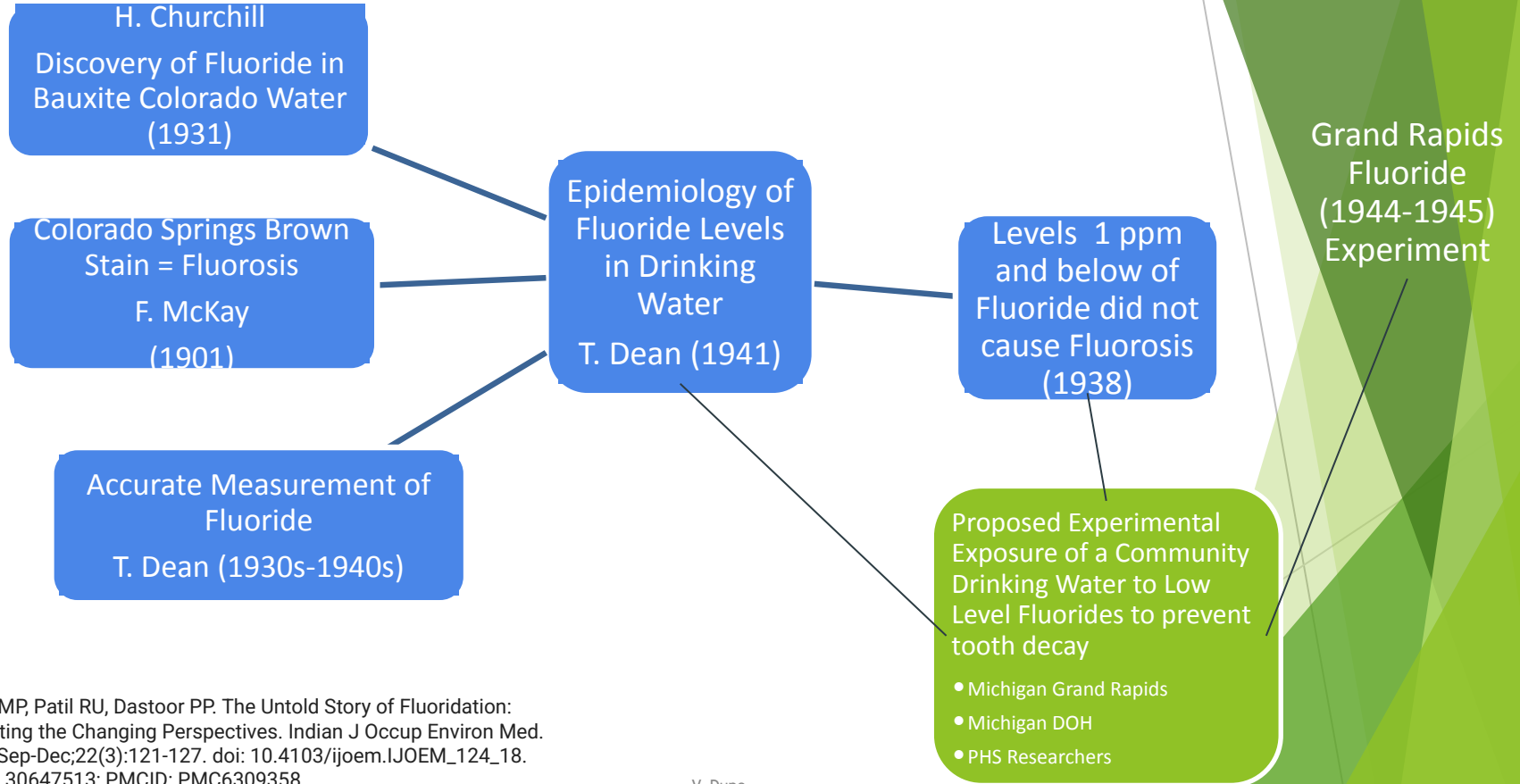
What is Research and Development?

- Development (progressionem)
 - As the latin implies this is an activity that considers moving forward with something into the future.
 - Development is working towards an end product or deliverable.

What is Research and Development?

- Research and development
 - Activity that engages engineers or/and scientists on a path to develop technology, process or system that can be either a service or product in the future.

What is Research and Development?

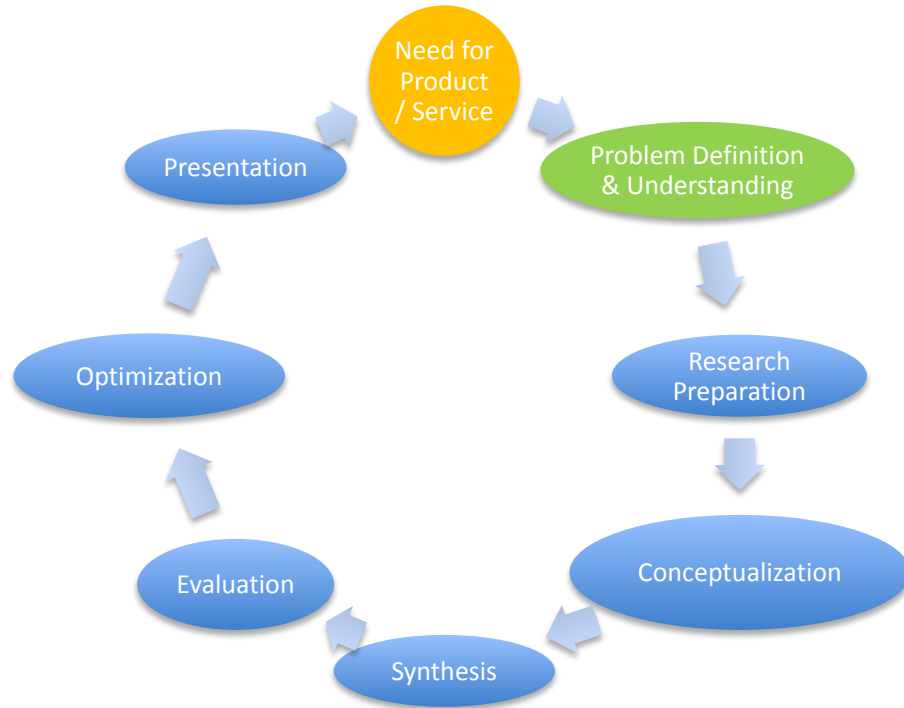


Unde MP, Patil RU, Dastoor PP. The Untold Story of Fluoridation: Revisiting the Changing Perspectives. Indian J Occup Environ Med. 2018 Sep-Dec;22(3):121-127. doi: 10.4103/ijoem.IJOEM_124_18. PMID: 30647513; PMCID: PMC6309358.

Engineering Design Process

The background of the slide features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green. These shapes are concentrated on the right side of the slide, creating a modern, dynamic feel. The left side of the slide is mostly white, providing a clean space for the text.

Basic Design Process



Basic Design Process



- Is there a real need for this?
 - Why?

Basic Design Process



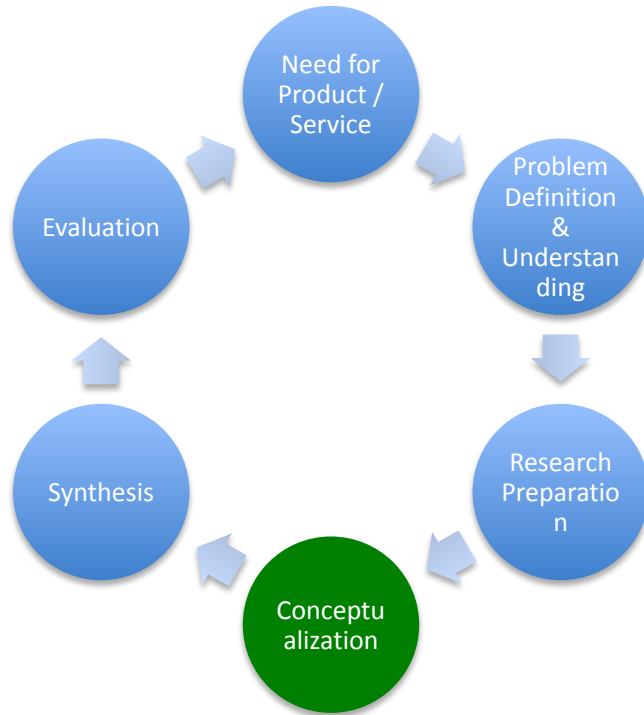
- What does it really solve?
- How does it solve it?

Basic Design Process



- Why do we need to solve it?
- Who benefits from it?

Basic Design Process



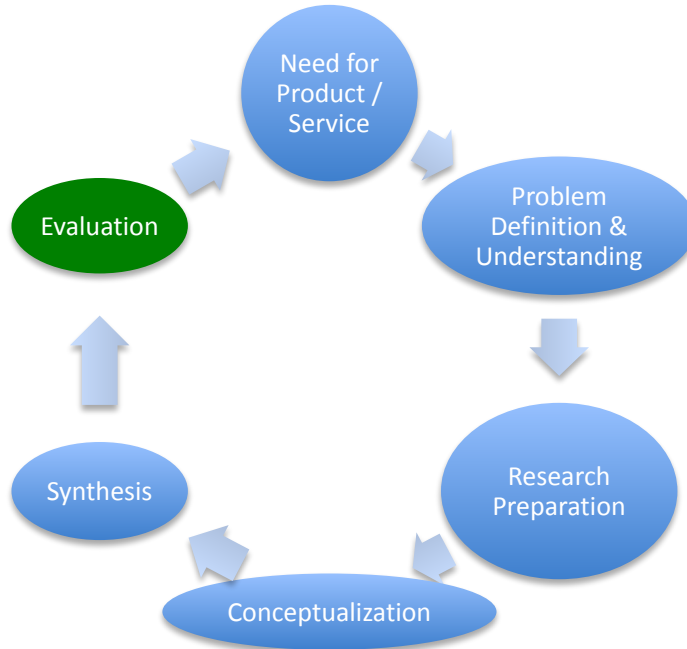
- What solutions can you create to address the problem?
- What do you need to create these solutions?
- How do you merge components to come out the the ideal solution?
- How much will it cost?

Basic Design Process



- Figuring out what works as viable solutions to the problem.
- Granted limit in
 - Materials
 - Time
 - Manpower
 - Funding

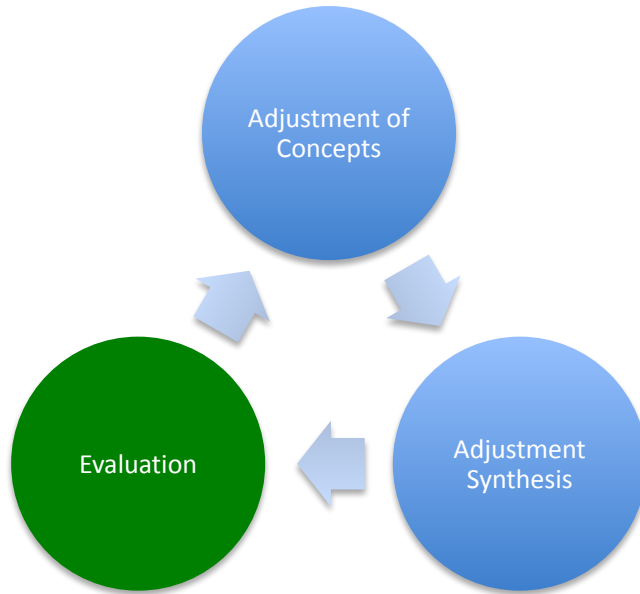
Basic Design Process



- Does it really do the task set?
- How do you go about testing it?
 - Accuracy?
 - Precision?

Basic Design Process: Optimization

- How can its performance be improved?



Design Process Final Step

- Presentation

- Problem
- Solution
- Capacity of Solution
- Cost
- Effectively of a solution

- Examples

- Product Pitch
- Conference Presentation
- Journal Publication

- Other Examples

- Demonstration of Product

Summary of Engineering Design Process

Stage	Area	Subjects
Needs Establishment	Specific to Requirement	Social Sciences, Current Affairs
Problem Definition and Understanding		English / Filipino Interview Process or Reading of Related Literature
Preparation Research	General Body of Knowledge	Reading areas related to the problem as past solutions or current ones
Conceptualization Synthesis	Specific to Area of Expertise	Professional Subjects
Evaluation	Testing Process	Statistical Design of Experiments
Optimization	Specific to Area of Expertise	Professional Subjects
Presentation	Venue to Present	English Communication Skills

Other Elements in Design Process

- Standards Enforcement (Engineering Management)

- 6. Controlling
 - 6.1. Control Systems
 - 6.2. Control Process
 - 6.3. Types of Organizational Control Systems
- 7. Managing Product and Service Operations
- 8. Managing the Marketing Function

Other Elements in Design Process

- Planning and Budgeting (Technopreneurship

1. Introduction

- Entrepreneurial Mindset
- Innovation and Ideas
- Products and Services
- Team Formation

2. Customers

3. Value Proposition

4. Market Identification and Analysis

5. Creating Competitive Advantage

6. Business Models

7. Introduction to Intellectual Property

8. Execution and Business Plan

9. Financial Analysis and Accounting Basics

10. Raising Capital

11. Ethics, social responsibility, and Globalization

& Engineering Management)

9. Managing the Finance Function

Other Elements in Design Process

- Staffing (Engineering Management)

- 4. Leading
 - 4.1. Leadership Across Cultures
 - 4.2. Sources of Power
 - 4.3. Leadership Models
 - 4.4. Kinds/Types of Leaders
- 5. Organizing
 - 5.1. Organizational Design
 - 5.2. Determinants of Structure
 - 5.3. Job Design and Job Characteristics Model

Other Elements in Design Process

- Audit Process for Administrative Area

(Engineering Management)

1. Evolution of Management Theory
2. Management and Its Functions
3. Planning
 - 3.1. Mission and Vision
 - 3.2. Stages of Strategic Planning
 - 3.3. Strategy Formulation
 - 3.4. SWOT Analysis, PEST Analysis and Porter's Five Forces Model
6. Controlling
 - 6.1. Control Systems
 - 6.2. Control Process
 - 6.3. Types of Organizational Control Systems
7. Managing Product and Service Operations
8. Managing the Marketing Function
9. Managing the Finance Function

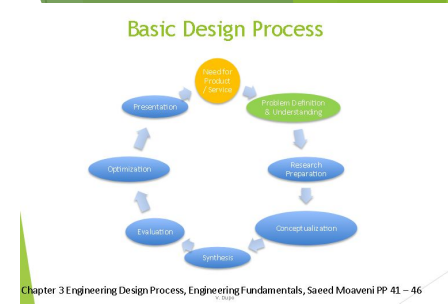
(Technopreneurship)

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8. Managing the Marketing Function

Why it's difficult to change this approach?

Reason is simple:

1. Academics is focused on Research not Research and Development.
2. The aim of studies equip the students with information, concepts and tools to work the problem and give a solution.
3. Evaluation system focuses on core concepts not preparation to work.



Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Reason is simple no human centered based Design Philosophy integrated into the curriculum

Stage	Typical Student Behavior	Characteristics / Common Issues
1. Problem Selection	Start with a technical problem (e.g., “build a robot that can climb stairs”) or a course requirement rather than a user-identified need.	- Driven by <i>technical challenge</i> , not <i>human need</i> . - Often framed around available tools or faculty suggestions. - Rarely validated with real users.

Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Reason is simple no human centered based Design Philosophy integrated into the curriculum

Stage	Typical Student Behavior	Characteristics / Common Issues
2. Concept Ideation	Brainstorm designs focused on mechanical or electronic feasibility — what's possible given their skills or materials.	- Solutions emerge from <i>engineering curiosity</i> rather than empathy. - Lacks exploration of multiple perspectives. - “Solution-first” mindset (they jump straight to building).

Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Reason is simple no human centered based Design Philosophy integrated into the curriculum

Stage	Typical Student Behavior	Characteristics / Common Issues
3. Design Development	Focus on technical specifications , drawings, and simulations (CAD, circuit design, structural calculations).	- Prioritizes efficiency, precision, and technical elegance. - Neglects usability, comfort, or accessibility. - Aesthetics and user interaction are secondary or ignored.

Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Reason is simple no human centered based Design Philosophy integrated into the curriculum

Stage	Typical Student Behavior	Characteristics / Common Issues
4. Prototyping	Build a single prototype (often the “best” idea from Step 2) for functionality testing.	- Prototype aims to <i>prove concept works</i> , not to <i>learn from users</i> . - Built with lab materials; rough ergonomics. - Iteration occurs only if there's a major failure.

Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Reason is simple no human centered based Design Philosophy integrated into the curriculum

Stage	Typical Student Behavior	Characteristics / Common Issues
5. Testing and Validation	Conduct technical tests (stress test, power consumption, accuracy).	- Tests confirm whether <i>it works</i> rather than <i>if it works for someone</i>. - No usability or human-centered evaluation. - Metrics are purely performance-based.

Evaluation

- Human Testing / Animal Testing
 - Protocol / Rules on how its done
 - Objectives
 - Removal of Bias
 - Data Handling and Interpretation
 - Informed Consent (Human Testing Only)
 - Liability and Risk Reduction (Human and Animal Testing)

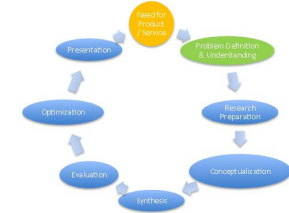
Why does Engineering Thinking in Design not do so well in preparing students for the industry?

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Stage	Typical Student Behavior	Characteristics / Common Issues
6. Presentation / Documentation	Submit final report or defense focusing on engineering rigor — calculations, efficiency, and performance graphs.	- Emphasis on <i>functionality</i> and <i>innovation from a technical angle</i>. - Success judged by performance, not adoption or satisfaction. - Users mentioned only in “potential applications.”

Why does Engineering Thinking in Design not do so well in preparing students for the industry?

Basic Design Process



Chapter 3 Engineering Design Process, Engineering Fundamentals, Saeed Moaveni PP 41 – 46

Stage	Typical Student Behavior	Characteristics / Common Issues
6. Presentation / Documentation	Submit final report or defense focusing on engineering rigor — calculations, efficiency, and performance graphs.	- Emphasis on <i>functionality</i> and <i>innovation from a technical angle</i>. - Success judged by performance, not adoption or satisfaction. - Users mentioned only in “potential applications.”

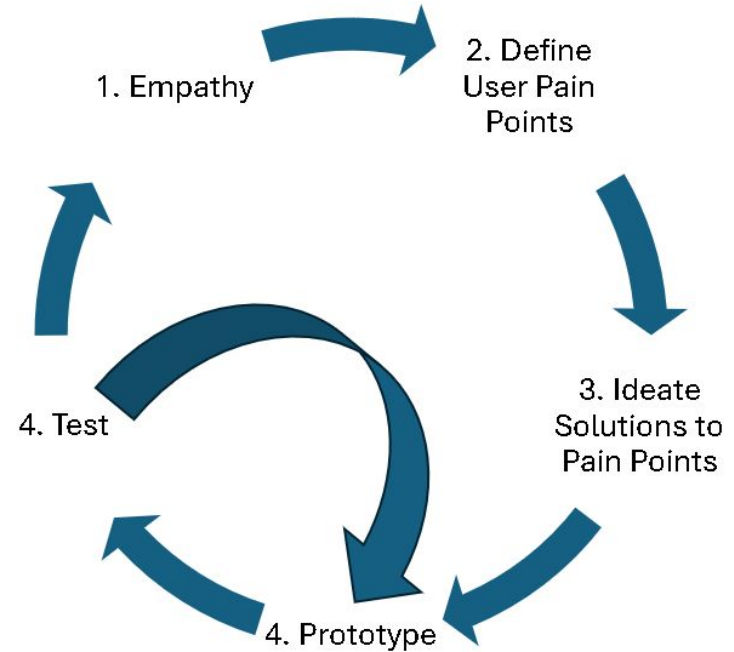
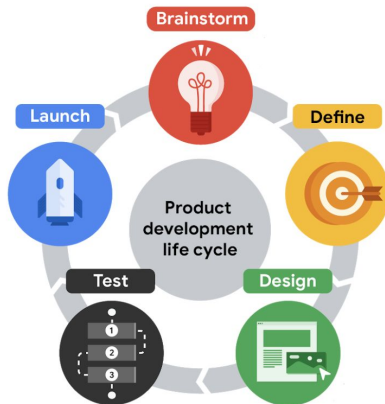
How the industry handles the problem?

- Engineering Cadetship Programs - Internal Training Programs
- Engineering Bootcamp with coordination with aligned academic institutions

Advanced Ideas in Design Process

- Design Thinking

- The inability of a design to consider human nature and human behavior the product ability to access the market or to be considered by the market as usable.



User research



Information architecture



Wireframing



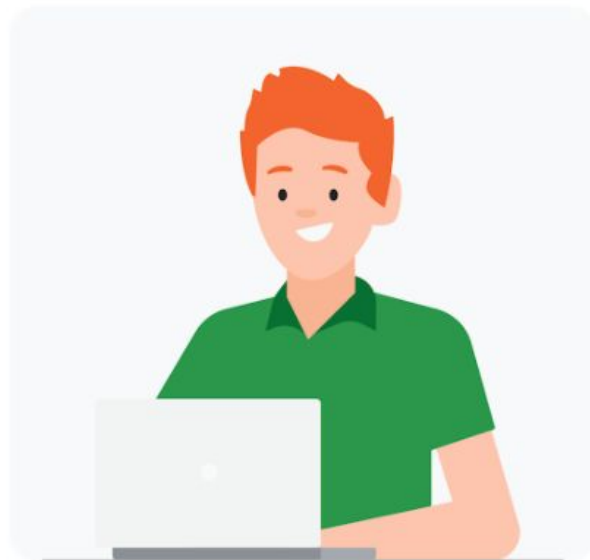
Prototyping



Visual design



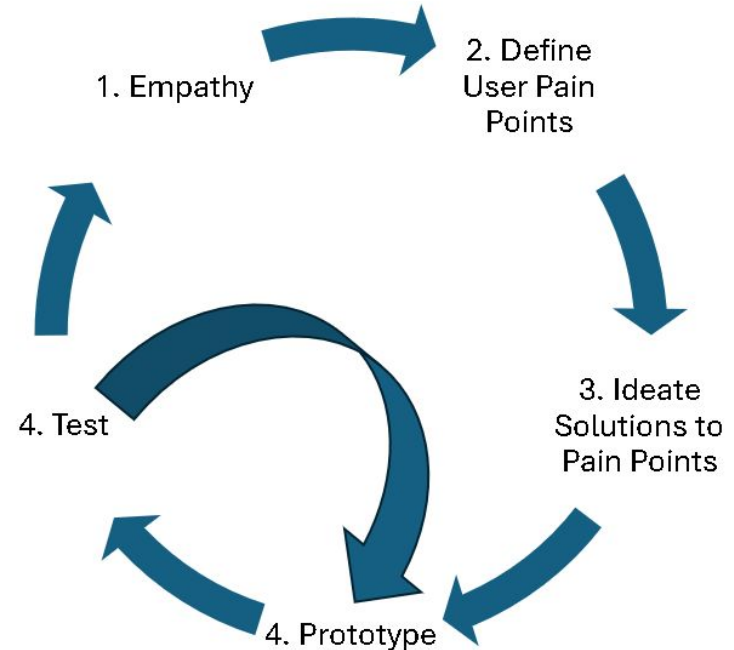
Effective communication



Alternative Design Process

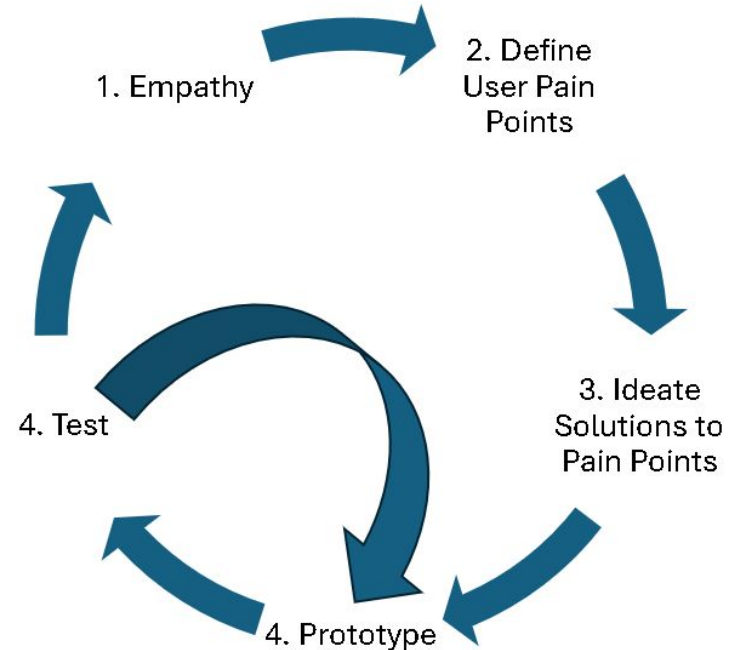
- Empathy

- Place yourself in the shoes of the end user.
 - ▶ Define what you need
 - ▶ What is currently being used?
 - ▶ How is current system failing?
 - ▶ Why do you think it is failing?
 - ▶ How do you feel about all of this?



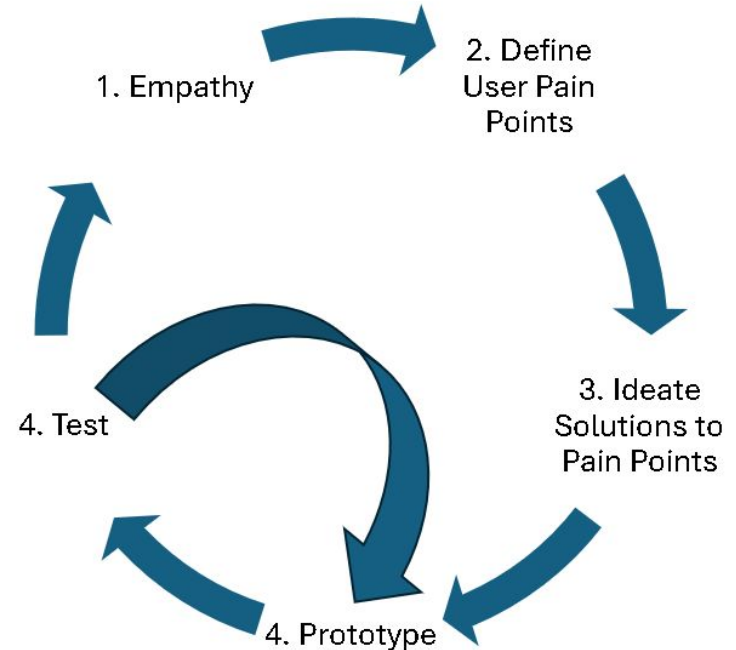
Alternative Design Process

- Empathy (Primary Tool - Interviews)
 - Talk to people and ask your questions.
 - ▶ Define what you need
 - ▶ What is currently being used?
 - ▶ How is current system failing?
 - ▶ Why do you think it is failing?
 - ▶ How do you feel about all of this?



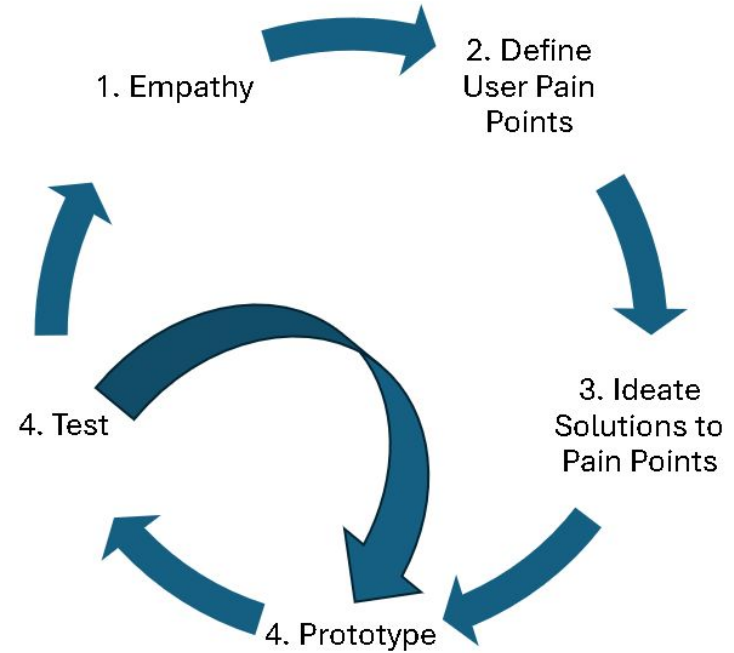
Alternative Design Process

- Empathy -> User Research
 - Place yourself in the shoes of the end user.
 - ▶ Define what you need?
 - ▶ Specifications
 - ▶ What is currently being used?
 - ▶ Competitor Audit
 - Pain Points defined
 - ▶ How is current system failing?
 - ▶ Why do you think it is failing?
 - ▶ How do you feel about all of this?
 - ▶ What is missing with current set of solutions



Alternative Design Process

- Empathy -> Personas
- Define your market into separate groups defined by what they represent
 - List out common themes of how each people think



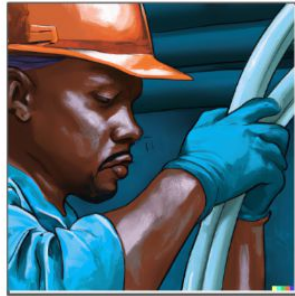
Alternative Design Process

- Empathy -> Personas
- Define your market into separate groups defined by what they represent

Persona: Lance

Problem statement:

Lance is a Certified Electrician who needs a way to directly see plans, request for inspections and report issues without trying to figure out where the engineers are and how to contact them because he just wants to finish his work on time with minimum interruptions.



Lance Oriel

Age: 25
Education: EIS NC II
Hometown: Bangued, Abra
Family: Single
Occupation: Electrician

“Work, Eat, Pray and Play life happens in between”

Goals

- Earn enough experience to begin his own company or to qualify for a job when he applies for migration to Australia.

Frustrations

- He would like to start his own business but the engineering management, accounting aspect of it all prevents him from even trying to learn it.
- He understands his disorganization is also limiting his possible promotion.

Lance is the only child of farmer. He moved to Metropolitan Manila 8 years ago. Sends money back to his folks every month. Lance works for an electrical construction company during 8am - 5pm but also has side jobs doing work outside for clients who need repairs in their installations. Lance likes doing the job in terms of using his hands. He follows rules but doesn't think too much about the management side like keeping receipts. He also is a bit disorganized.

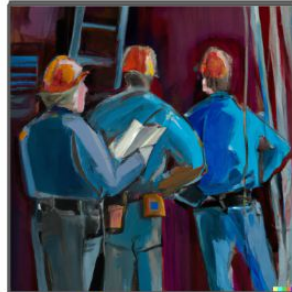
Alternative Design Process

- Empathy -> Personas
 - Define your market into separate groups defined by what they represent
 - List out common themes of how each people think

Persona: Marc

Problem statement:

Marc is a contracted services business owner who needs a simple structured means to document and report costs from each site to my management team because he wants to grow his business without spending too much time to train new engineers on that data handling and administrative part of project management..



Marc Reganit

Age: 45
Education: Electrical Engineering
Hometown: Laurel, Batangas
Family: Married with Kids in College
Occupation: Professional Engineer & Contractor

"Even as technology and science progresses if the students are not able to catch up it causes significant problems in day to day operations"

Goals

- Increase projects handled by four times during building boom

Frustrations

- Has problems overseeing multiple projects since the quality of junior engineers documentation of expenses making it difficult to show the value of his companies work done.

Marc Reganit learned at 23 how to manage projects in electrical engineering. He started his own company at 25 and has been working to find ways to promote his company that provides good quality work for an affordable price. The last 25 years he has seen a decline in the quality of his younger engineering staff to keep tabs on materials, costs and work done. He wants his company to be more unique in the way it interacts with clients and the data their projects generate.

Alternative Design Process

- Empathy -> Personas
- Define your market into separate groups defined by what they represent
 - List out common themes of how each people think

Persona: Rika

Problem statement:

Rika is a busy paramedic, single parent who needs a means to see how her renovation projects are progressing as per plan and budget by simply looking at my phone and not spending time going over stuff that she doesn't understand because she wants to have more time to take care of my daughter, continue on my changing work shifts and still be able to rest my body and mind.



Rika Yamakawa

Age: 29

Education: EM Services NCII

Hometown: Pila, Laguna

Family: Widow with daughter

Occupation: Paramedic

"Keeping tabs on renovations and costs is another added work while living in a 150 year old house balancing that with my work shifts and taking care of my daughter"

Goals

- To be able allow daughter to be educated until College Level.
- Wants renovate her home to make it into a AirBnB for Tourists

Frustrations

- Rika sa issues with reading text of a specific size and font.
- Finds the renovation process of her family home a very expensive and confusing to track.

Rika is a paramedic who lives in Pila. She lives with her daughter. Her husband died 7 years ago. Rika works both day and night shifts. The work schedule depends on the availability of other paramedics. Both she and her daughter like the outdoors. Rika suffers from blind spots in her peripheral vision as a result of laser surgery making it difficult for her to read small font size text. Rika believes that being health-conscious is partly watching what she eats along side constant exercise. The house she lives in currently is 150 years old she has been slowly upgrading the wiring and fixtures.

Alternative Design Process

- Empathy -> Journey Map
- Ask your users what they do normally in relation to the problem area. Go thru each step find out how they approach the task and how they feel about it. Do not comment only listen and take notes

Persona: Marc Reganit

Goal: Monitor Material and Work done on site

ACTION	Staff scheduling	Deploy Staff	Retrieve Data	Process Material Type, Quantity and Project Status	Prepare Site Reports for Clients	Brief Clients
TASK LIST	1. Check Staff Availability 2. Organize Engineering Staff assignments per week 3. Tell engineers of their deployment details for Monday	1. Let staff report to sites 2. Ask them to collect receipts and work hours rendered by electricians. 3. Ask them to note quality or quantity issues	1. Ask staff to return data on Thursday afternoon 2. File receipts to the main office by site	Ask Engineers on Friday with Office staff to pour into receipts to categorize it by type, record its costs, section installed at. Store processed receipts in company records.	1. Write the running costs vs progress on worksite. 2. Write briefing data for clients	1. Wait for Clients to request for an update via sms/phone call/email 2. Set appointment for meeting if client requests a personal appearance for the briefing
FEELING ADJECTIVE	1. Anxious about the proper assignment of staff to the work sites and accuracy of staff availability 2. Settled that this is the best staff schedule that can be done.	1. Relieved that supervisory staff is on site 2. Anxious if receipts and work hour notes have been misplaced during the last week one day return.	Concerned that staff may be returning too early	Anxious that engineers are not supervising while back in the office and receipts may get lost	Anxious that engineers are not supervising while back in the office	Stressed about fitting schedules between clients
IMPROVEMENT OPPORTUNITIES	Can staff also use the application to file their reporting availability.	A system has to be setup guarantee the proper manage receipts on site and store them while supervising engineers are outside the site.	Return to office schedules have to be reworked.	Have a way to obtain receipt information for materials and work done even if there are no supervising engineers.	Make time spent by engineers away from project sites lesser.	Reduction of stress about working out a schedule to fit clients.

Alternative Design Process

- Empathy -> Journey Map
- Ask your users what they do normally in relation to the problem area. Go thru each step find out how they approach the task and how they feel about it. Do not comment only listen and take notes

Persona: Rika Yamakawa

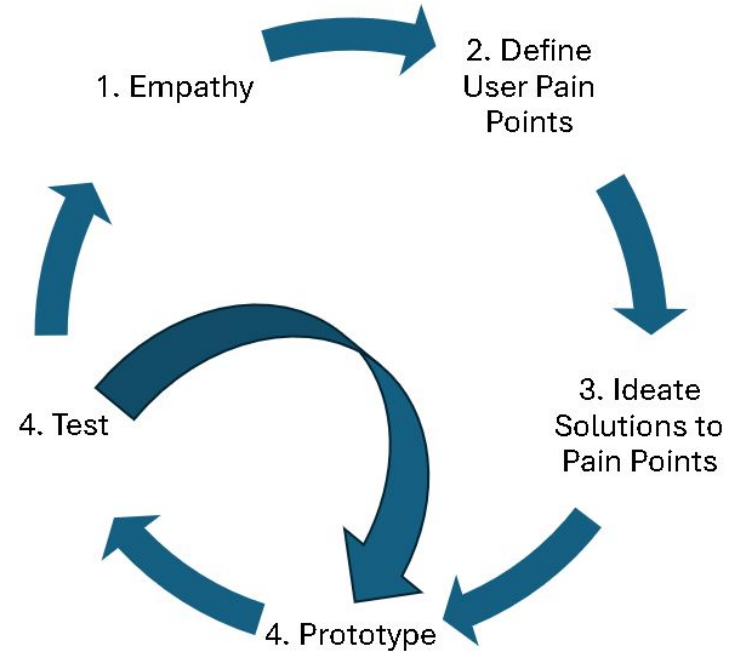
Goal: Monitor Progress of Project and accumulated costs

ACTION	Call Contractor about project update	Set Schedule day of meeting with contractor	Meet with contractor
TASK LIST	1.Message the contractor about updates on the project work or proposals. 2. List possible days for the meeting.	1. Receive reply on confirm or reset the schedule for the update.	1.Proceed to set meeting area. 2.Listen to the presentation done by the contractor 3.Ask questions.
FEELING ADJECTIVE	1. Settled that an update meeting is needed. 2. Hoping the set days available will coincide with days the contractor is free.	1.Anxious if the set schedule will work out.	1.Stressed Travelling away from home or waiting for them to arrive. 2. Confused about the technical Terminology of the reports. 3.Anxious about cost increases and delays.
IMPROVEMENT OPPORTUNITIES	1. Reduction of anxieties regarding meetings.	1. Reduction of anxieties regarding meetings.	1. Reducing confusion jargon used in the reports. 2.Reduce stress created by travelling or hosting visitors.

Alternative Design Process

- Define User Pain Points

- The persona frustrations and journey maps improvement opportunities allow us to glimpse the User Pain points we can set ourselves to address with our device.



Value Proposition

Is a summary of why a consumer should use a product or service.

Step 1. Describe your product's features and benefits. Create a list of all the great features and benefits of your product, big and small. Don't hold back; list everything that comes to mind and then narrow it down later.



Step 2. Explain the value of the product. Anything that you identify as a value proposition needs to be beneficial to your users. In this example, for the dog walker app, there were four categories of product values that were identified during user interviews: accessible, professional experience of the dog walkers, cost, and reliability.



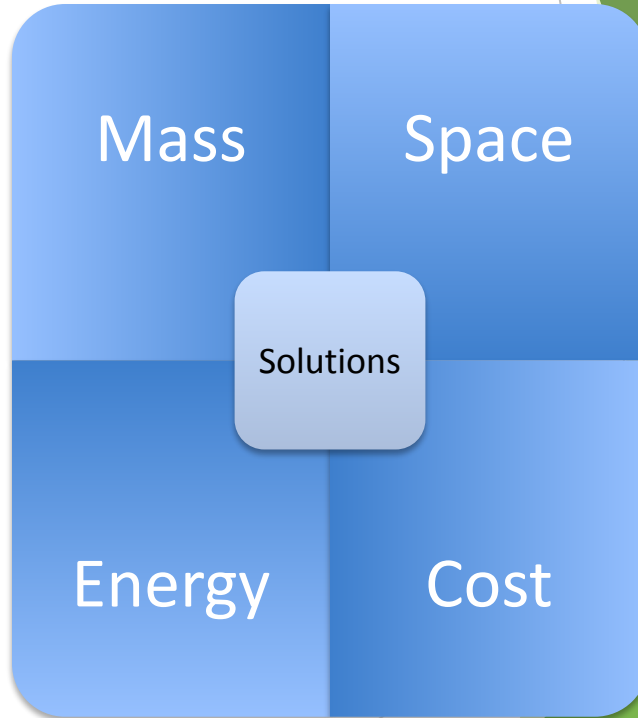
Another way to help define Value Proposition

Instructors Recommended Approach

- Teoriya Resheniya Izobretatelskikh Zadatch (TRIZ)

— The ideal solution is one that:

- Occupies no Space
- Has no Mass
- Consumes no Energy
- Costs Nothing



Example: Writing data results in micro-gravity environment

NASA USA Solution

- Cost: 1M USD Supposedly
- Technology:
 - Hermetically Sealed and Pressurized Ink Well and ball point tip
- Works in Micro and Zero G Environments
- Mass = 20g

USSR Solution

- Cost: 0.55 Cents
- Technology
 - Graphite Encased in two laminated wood panels. Needs Sharpening only
- Can work in Micro-Gravity and Zero Gravity Environments and wet environments.
- Mass 20g

Example: Writing data results in micro-gravity environment

NASA USA Solution



USSR Solution



Example: Writing data results in micro-gravity environment

Energy: Melt and Form Metal
Pressurize and Seal Vessel



Energy: Make rubber, Metal Forming and Melting,
Wood and Graphite Cutting and Lamination



Step 3. Connect these features and benefits with the needs of your users. The goal is to identify what's truly valuable to the user and not just a cool feature that users didn't ask for. To determine value, take the personas you've developed and pair each persona with a value proposition that meets their biggest pain point.



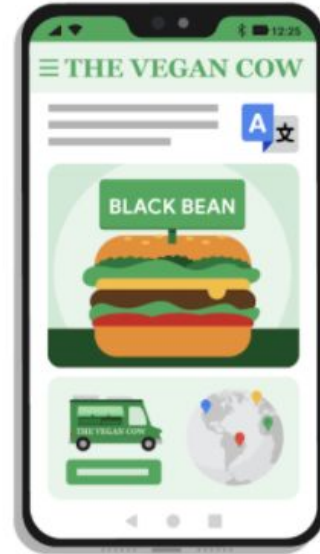
Competitive Audit



Your product

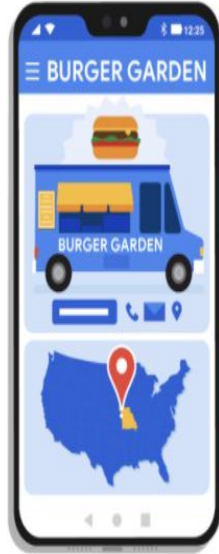


Direct competitor



Indirect competitor

Competitive Audit



Your product



Direct competitor



Indirect competitor

Competitive audits can offer many benefits throughout your ideation phase, including:

- Giving you an idea of products already in the market and their designs.
- Suggesting ideas to solve early problems that you're facing with your own designs.
- Revealing the ways that current products in the market are not meeting users' needs. This is a gap for your product to address!
- Demonstrating the expected life cycle of a product in the same market as yours.
- Informing all the different iterations your product could take and how those performed for your competitors.

Competitor Audit

Part 2 - Competitive Audit Report

Google UX Design Certificate

1. Competitive audit goal(s)

Comparing the overall experience in cost and reporting for applications that can work with electricians

2. Who are your key competitors? (Description)

Easy Expense, Expensify are our direct competitors with indirect ones are Expense Manager and Monday.com

3. What are the type and quality of competitors' products?

(Description)

Easy Expense, Expense Manager and Expensify offer receipt information acquisition that helps populate the cost database. Reports are generated from this area. Expense Manager adds to this by adding running totals and debt to income analysis. Expensify adds to it by placing an communications option to allow users to cooperate and share information. Monday.com provides a way to follow up on progress for the project with users populating this requirement on their own.

4. How do competitors position themselves in the market?

(Description)

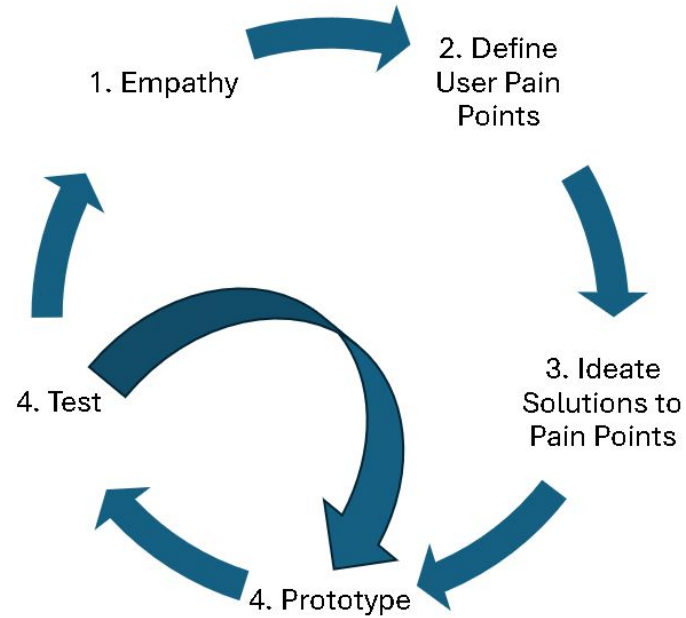
Easy Expense and Expensify provide OCR integration to the data collection process making it easier to obtain details and collect cost information. Expense Manager positions itself as a way to monitor costs for a team or a company. While Monday.com is positioned as an effective work coordination tool.

9. Opportunities (List)

- Create a product that allows for different user types to see an interface that is tailored to their needs.
 - It can be done this way since each one has different side to the story they seek to define.
 - This will reduce clutter and let people focus on the purpose of the app for them instead of bundling everything in one gui.
- Creation of three applications for the contractor side that are working together in the back end to provide a service for contractors that can:
 - Application 1
 - Receive Task List to execute for the day.
 - Account for Costs using receipts
 - Account for Labour Hours Rendered per segment of the project
 - Application 2
 - Assign and Update Tasks for more efficient work
 - Account for Materials and Inspection of Installation
 - Report Progress of Work
 - Application 3
 - See each projects progress and costs versus budget
 - A way to issue instructions and modify workflow to application 2 users without taking too much time.
- Creation of a fourth application that has a direct line to Application 3 users to obtain updates on projects, set schedules for meetings and see billing statements. This is aimed at the Client for the firm.

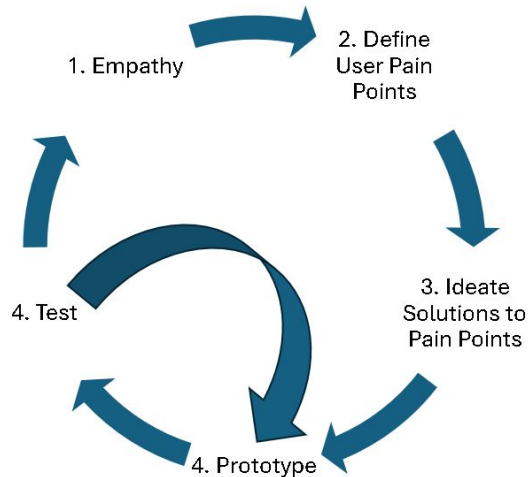
Alternative Design Process

- Ideate
 - Means to select a pain point to address.
 - Come up with a solution set that can remove or lessen the pain point.



Alternative Design Process

- Ideate
 - Means to select a pain point to address.
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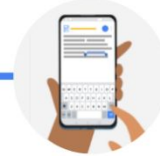


Alternative Design Process

- Ideation

Characteristics of a Good
User Experience

Usable



Equitable



Enjoyable



Useful



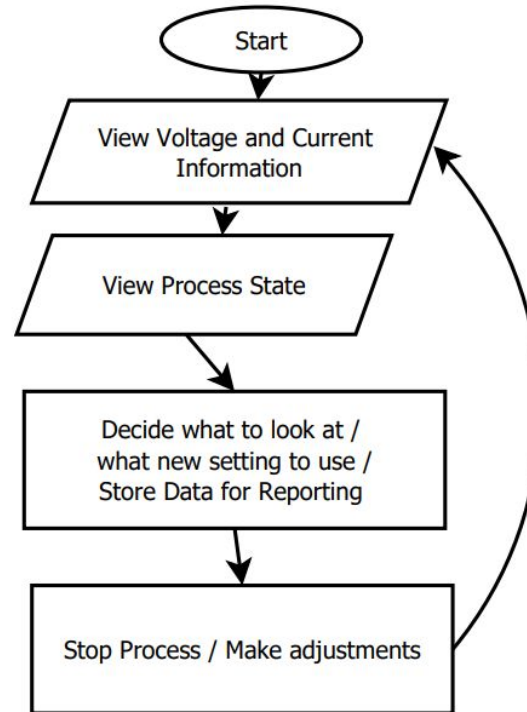
Alternative Design Process

Console Button Design



Alternative Design Process

Console Button Designs



Alternative Design Process

Information Architecture

Information Architecture (IA) is the practice of organizing, structuring, and labeling content and functionality in a clear, logical way to help users find information and complete tasks efficiently.

It's used across digital products (websites, apps, intranets, software) and includes:

- Defining navigation systems
- Categorizing and labeling content
- Structuring hierarchies (menus, sitemaps)
- Organizing workflows and user journeys

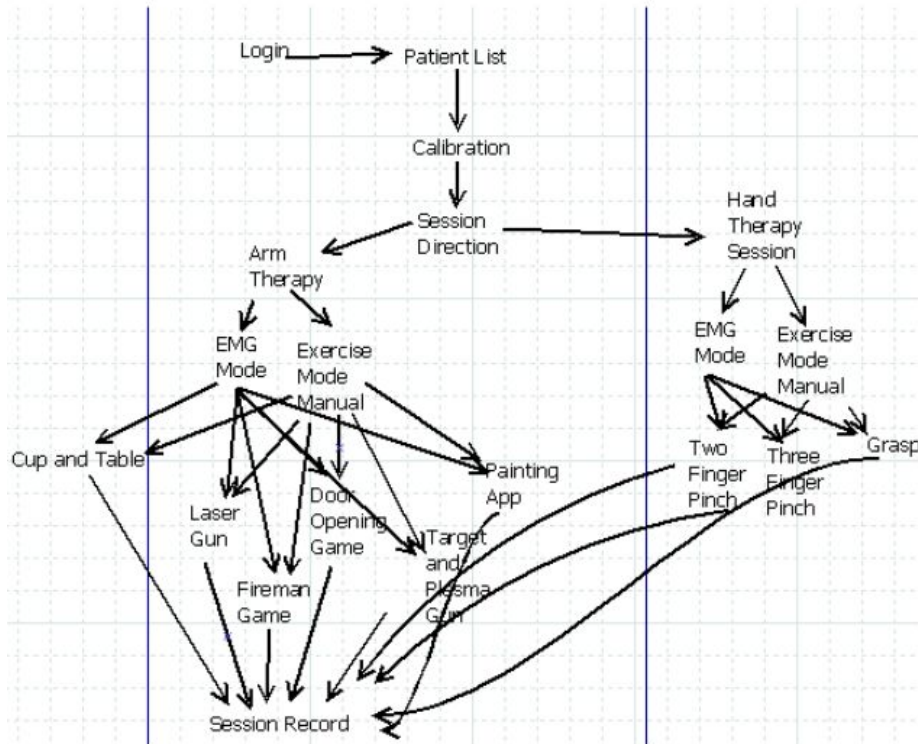
Goal: Make content findable, usable, and intuitive.

Information Architecture

I've observed that:

The clearer and shorter the information architecture is the easier it is for users to learn the operations.

Information Architectural Design

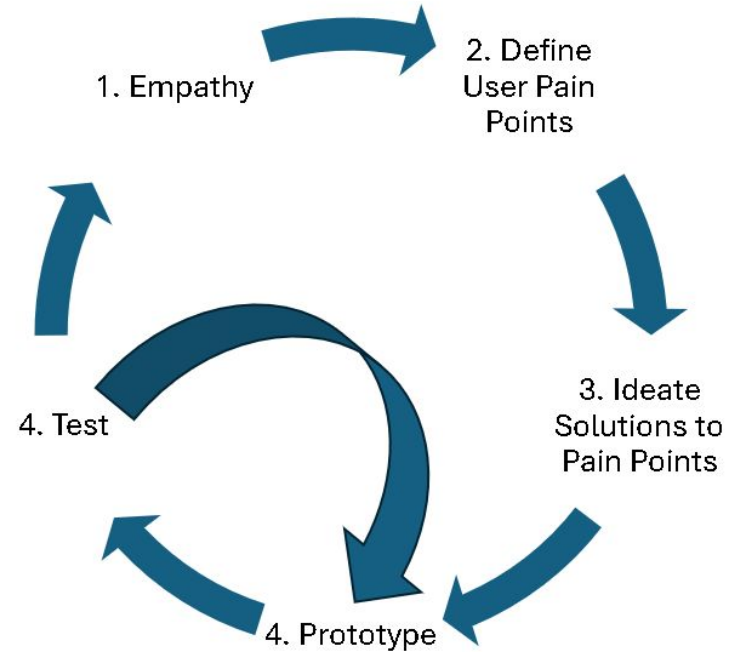
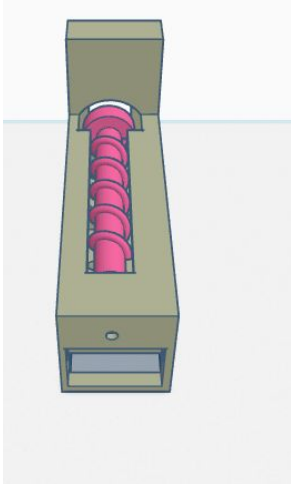


Issues

- Session Records are Electronic saveable thru CSV
- Patient Data shows too much information when listing
- Exercise Modes for the Arm assume whole arm exercises it should be targeted instead.
- Session Records show all patients this is an issue since it exposes too much data to the user.
- APRO interface integrated with RESTORE should be removed from this one.
- Version for PC and Version for Mobile
- Standard Data Container Format for Data sharing via Lan or Health Record System missing

Alternative Design Process

- Prototyping without doing fabrication or buying parts
 - Wireframing
 - 3D CAD Assemblies
 - Simulation Tools

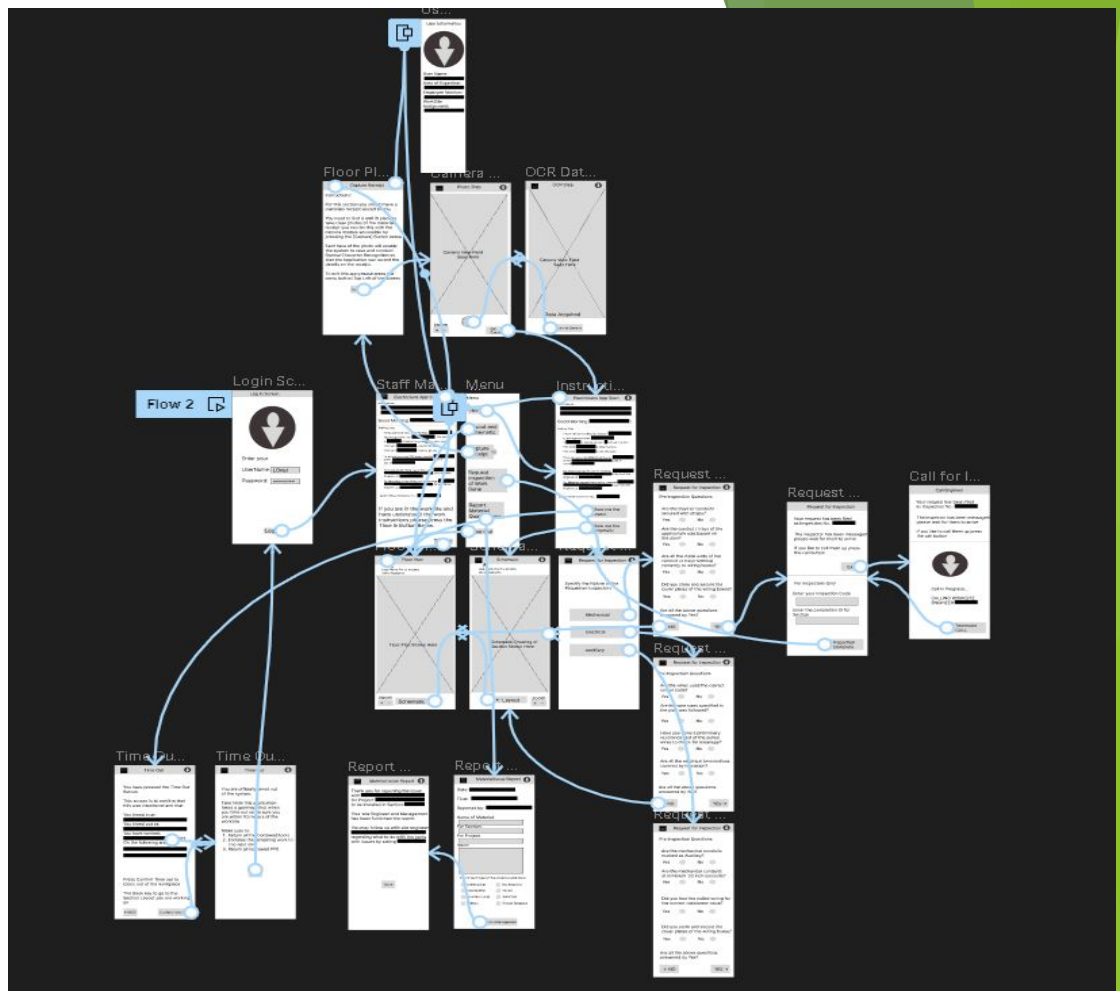


Low-fidelity prototype

This is the low fidelity prototype for the Electricians module for Lance

To test it out here is the link

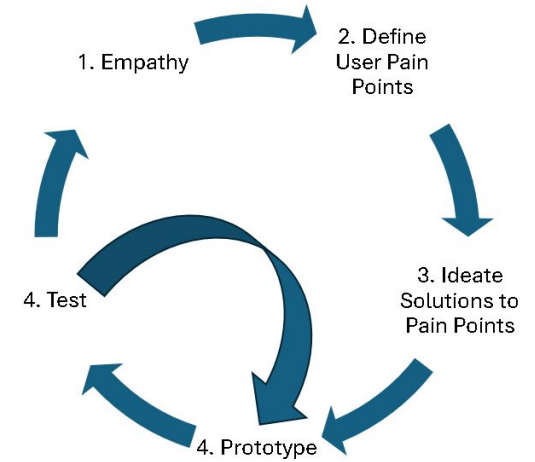
<https://www.figma.com/proto/7Seenh0nPh14O6qAgQBOWd/Electricians-Module-for-Electrician-Mobile-App?type=design&node-id=23-429&t=uyzrGyv53E1B0RBw-0&scaling=min-zoom&page-id=0%3A1&starting-point-node-id=23%3A429>



Alternative Design Process

- Prototyping without doing fabrication or buying parts

- Wireframing → Front End Interface Programming
- 3D CAD Assemblies → CNC/3D Printing
- Simulation Tools → Embedded Systems Programming

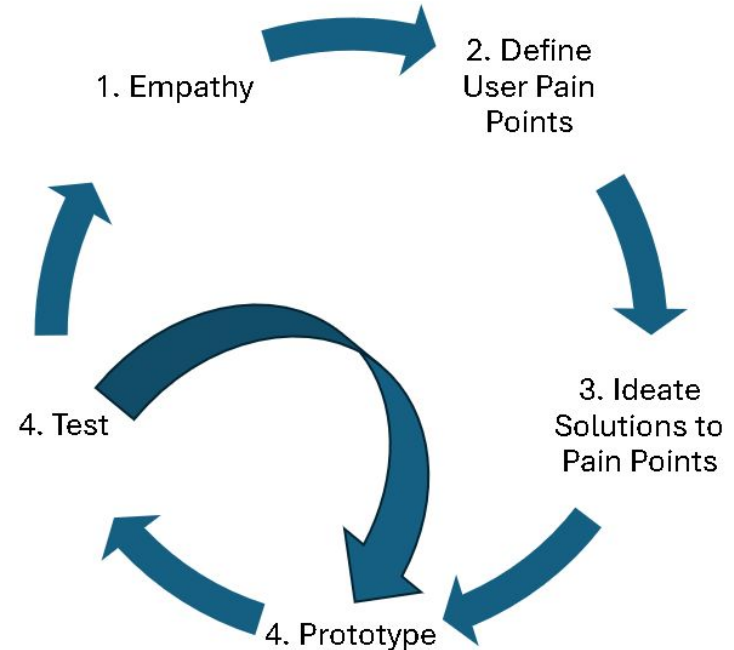


Advanced Ideas in Design Process

Instructors Recommendation not Part of Syllabus

- Testing

- Depends on what you are evaluating
- Wireframing
 - ▶ Actual Interface Test with humans
 - ▶ Ease to Navigate
 - ▶ Ease to Learn
 - ▶ Practicality of Design
- Full CAD Model
 - ▶ Human Perception of Style or implementation
- Simulation Tools
 - ▶ Actual Software Estimates on Performance

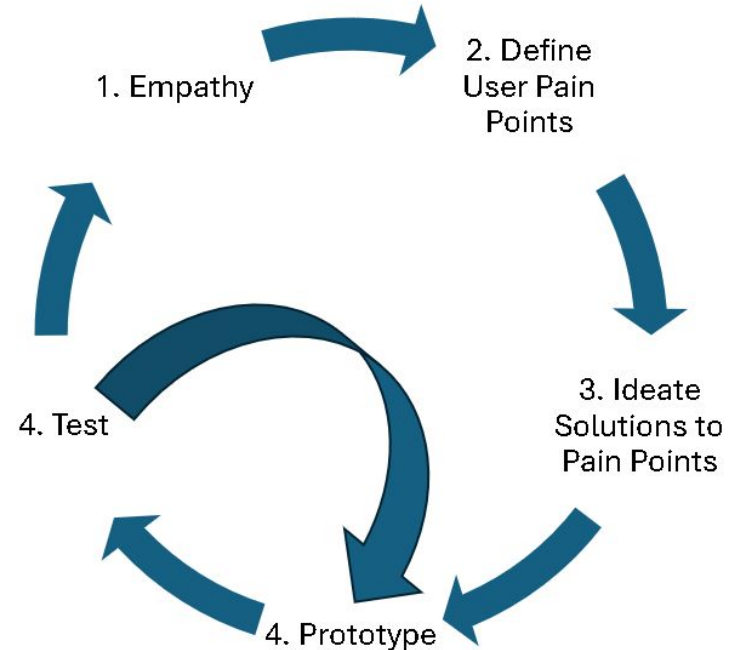


Advanced Ideas in Design Process

Instructors Recommendation not Part of Syllabus

- Design Thinking

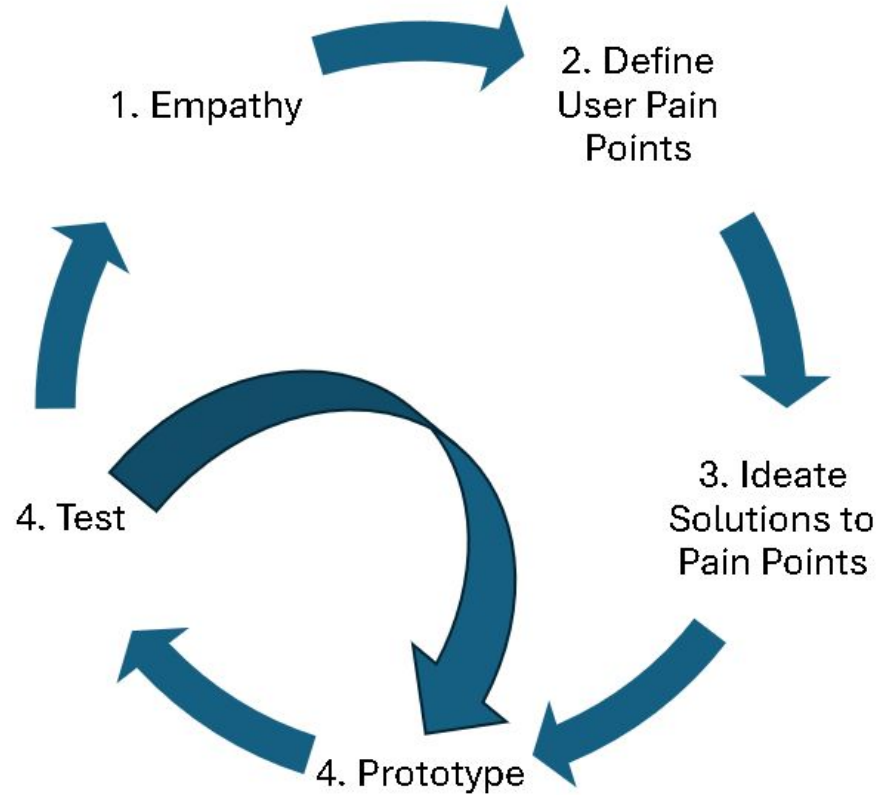
- The inability of a design to consider human nature and human behavior the product ability to access the market or to be considered by the market as usable.



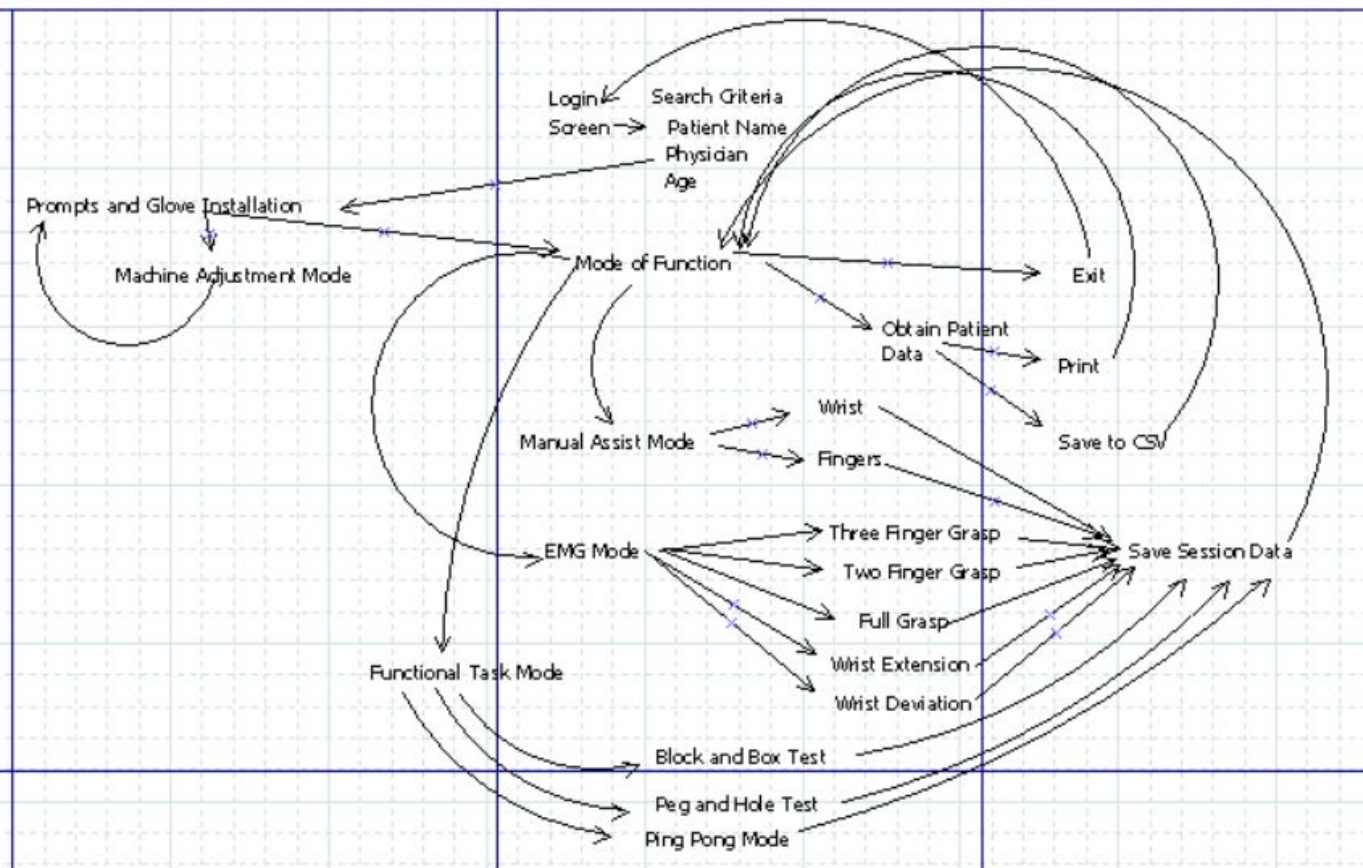
User Experience

User Experience is a study on the response of a people to a product / service measuring the level of desirability of their responses in terms of:

- Performance
- Interface
- Ease of Use
- Ease to Learn

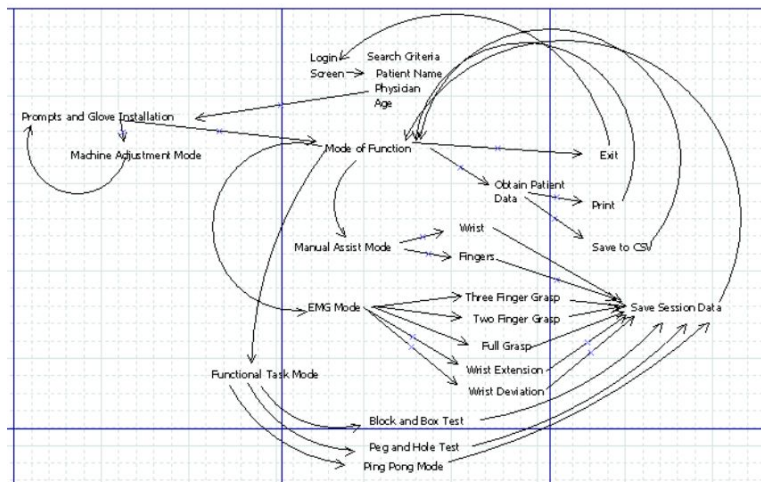


Information Architecture v2.0



CSV Format should comply
With national Standard
For data Sharing

Information Architecture v2.0



CSV Format should comply
With national Standard
For data Sharing



Revised Information Architecture

5:56 3%

This is an experimental software for APRO

Keypad Based Voice Based

IP Address 192.168.100.5

Listen Start 1234

Full Grasp 3 Finger Grasp

Fingers at Rest 2 Finger Grasp

Wrist Ulnar Deviation Wrist Radial Deviation

Wrist at Rest

Finger Exercise 3

Finger Exercise 2 Finger Exercise 1

Wrist D-V Wrist U-R

Repetition 1

Keypad Mode

voltaire.dupo@dlsu.edu.ph

x55

5:57 3%

This is an experimental software for APRO

Keypad Based Voice Based

IP Address 192.168.100.5

Listen Start 1234

Repetition 1

Google

Didn't catch that. Try speaking again.

Try again

English (Philippines)

Google speech services converts audio to text and shares the text with this app.

5:57 3%

This is an experimental software for APRO

Keypad Based Voice Based

IP Address 192.168.100.5

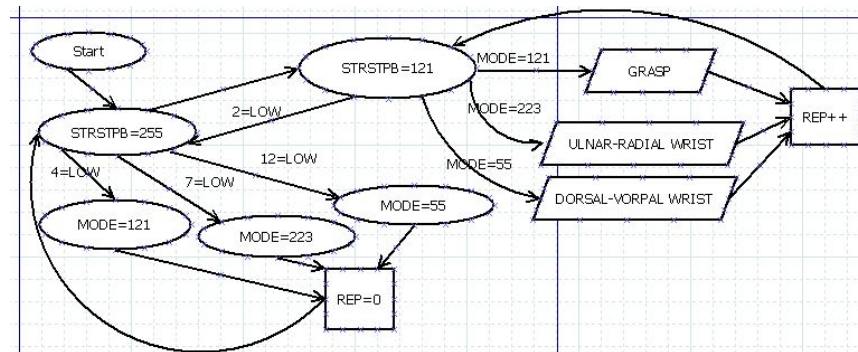
Listen Start 1234

Repetition 1

Voice Mode

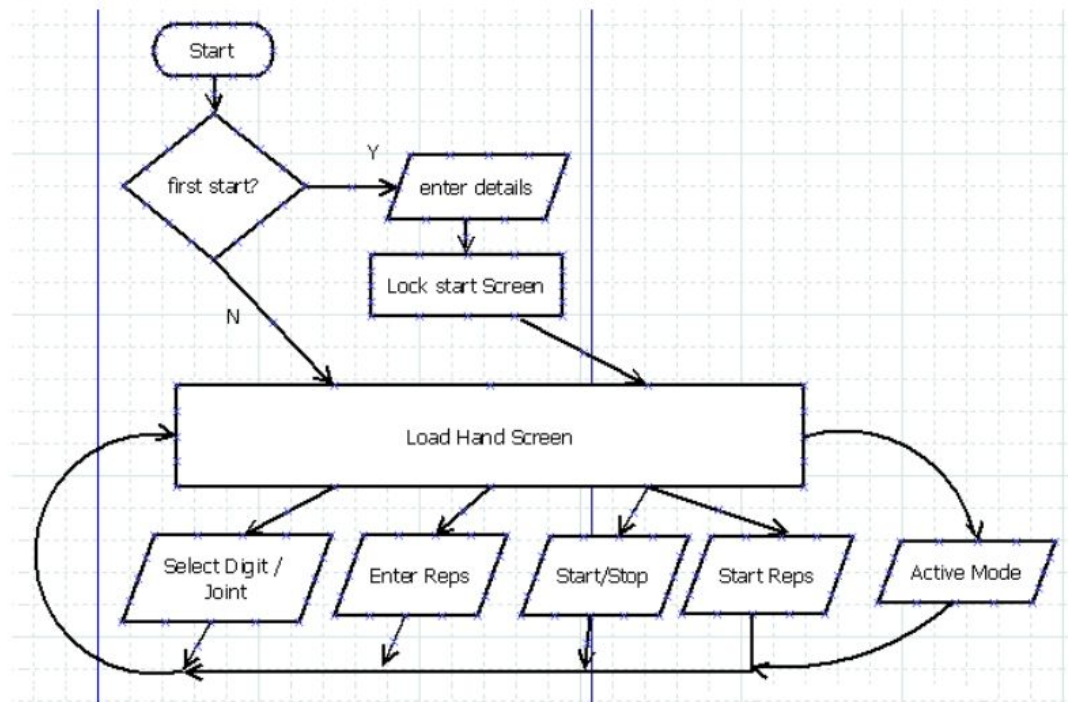
voltaire.dupo@dlsu.edu.ph

q150





Harry Bao's GUI for APRO Information Infrastructure



Hardware Design Sample

APRO Glove

Aira Patrice R. Ong's Glove Design

- Moves the digits to extend or flex
- Garter channels for the fingers



Summary of initial User Feedback

Recommendation is to fix issues and retest the mechanism

Round 0 findings

- 1 Locking Mechanism not too effective for securing cables
- 2 Takes around 15 to 30 minutes to address attaching cables.
- 3 No Tactile Sensing when Glove is worn
- 4 Difficult to fit hand into glove
- 5 Glove is too heavy
- 6 Glove is for right hand only no left hand version.

APRO Glove with Wrist Version 1.5

Covered by Patent in the Philippines

- Moves the digits to extend or flex and Garter channels for the fingers (Aira Ong)
- Moves wrist (Aira Ong and Voltaire Dupo)
 - Ulnar and Radial Deviation
 - Supination
 - Flexion
- Revised Knuckle Tie Block (JV Cortez)
- Tube Channel (Jacob Reyes)
- Shrinkable Rubber Tube for the Cable Channel / Guide



Usability study: findings for Revised Glove Testing

This is summary of findings for the glove

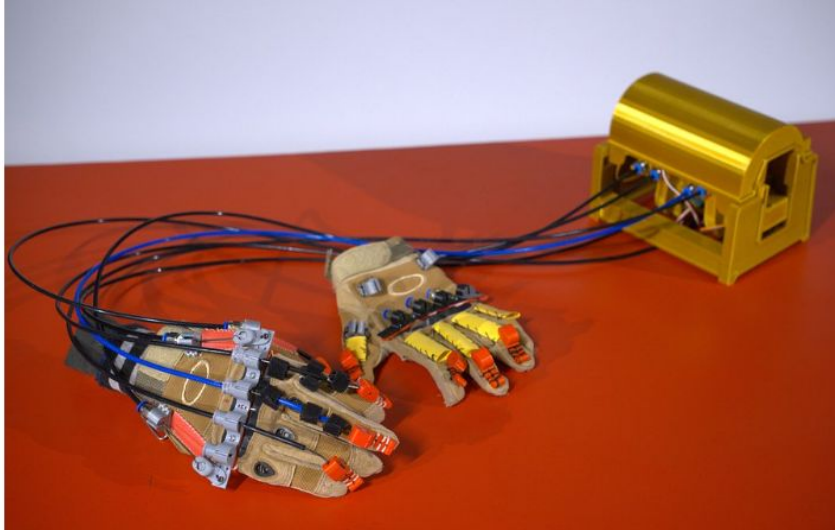
Round 1 findings

- 1 Finger Pad Slip Out Issue
- 2 Loose feeling on the palm area but the fingers fit snugly
- 3 Position of Finger Pusher Blocks have a tendency to dig into nails
- 4 Stress on Knuckles as a consequence of cable extending and pushing fingers to curled position.
- 5 Item 5
- 6 Item 6

APRO Glove with Wrist Version 2.0

Exhibited in CES 2025 in Las Vegas, USA

Primary Hardware Designers : Voltaire Dupo ,
John Vincent Cortez
Industrial Design: Sachi So.



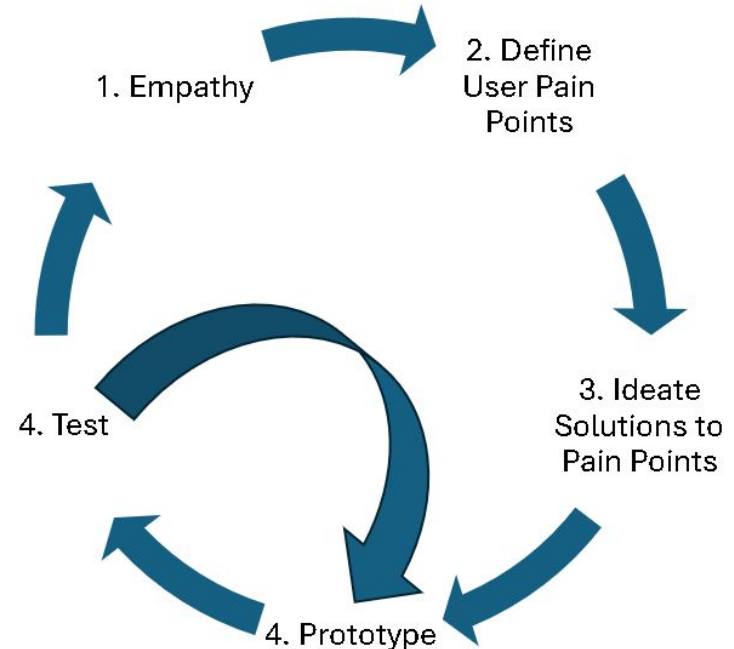
Alternative Design Process

- When is testing enough?
 - When the journey map no longer shows the same frustrations

Persona: Marc Reganit

Goal: Monitor Material and Work done on site

ACTION	Staff scheduling	Deploy Staff	Retrieve Data	Process Material Type, Quantity and Project Status	Prepare Site Reports for Clients	Brief Clients
TASK LIST	1. Check Staff Availability 2. Organize Engineering Staff assignments per week 3. Tell engineers of their deployment details for Monday	1. Let staff report to sites 2. Ask them to collect receipts and work hours rendered by electricians. 3. Ask them to note quality or quantity issues	1. Ask staff to return data on Thursday afternoon 2. File receipts to the main office by site	Ask Engineers on Friday with Office staff to pour into receipts to categorize it by type, record its costs, section installed at. Store processed receipts in company records	1. Write the running costs vs progress on worksite. 2. Write briefing data for clients	1. Wait for Clients to request for an update via smartphone call/email 2. Set appointment for meeting if client requests a personal appearance for the briefing
FEELING ADJECTIVE	1. Anxious about the proper assignment of staff to the work sites and accuracy of staff availability 2. Settled that this is the best staff schedule that can be done.	1. Relieved that supervisory staff is on site 2. Anxious if receipts and work hour notes have been misplaced during the last week one day return.	Concerned that staff may be returning too early	Anxious that engineers are not supervising while back in the office and receipts may get lost	Anxious that engineers are not supervising while back in the office	Stressed about fitting schedules between clients
IMPROVEMENT OPPORTUNITIES	Can staff also use the application to file their reporting availability.	A system has to be setup guarantee the proper manage receipts on site and store them while supervising engineers are outside the site.	Return to office schedules have to be reworked.	Have a way to obtain receipt information for materials and work done even if there are no supervising engineers.	Make time spent by engineers away from project sites lesser	Reduction of stress about working out a schedule to fit clients.

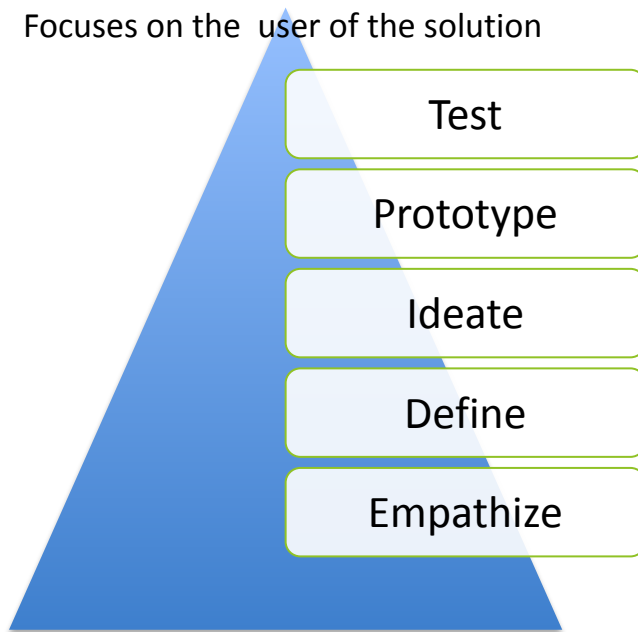


Advanced Ideas in Design Process

Instructors Recommendation not Part of Syllabus

Design Thinking

Focuses on the user of the solution



Engineering Thinking

Focuses on the solution to the Problem

